

Graduation Project Info 46 — Alexandria

Devotopia — Verified Skill Roadmaps — submission answers, question by question

Captured from the live form before submission. Figures come from the project source: plan prices from `payment.dto.ts`, the adaptive scoring rules from `assessment.service.ts`, the 30% remedial threshold from `remedial-node-queue.service.ts`, and the provider chain from `ai-provider.factory.ts`.

1. Project identity

Choose your track

Open Source

Project name

Max 80 characters. Printed in the graduation catalogue.

Devotopia — Verified Skill Roadmaps

Tagline

One line saying what the product is. A description, not a slogan.

AI career-preparation platform that turns a target job role into a personalised learning roadmap and verifies every skill with an adaptive exam.

2. Problem and solution

The problem

Min 300 characters.

Egyptian computer-science graduates and self-taught juniors enter a hiring market where every CV claims skills nobody has verified. A learner picks a target role such as Full-Stack Developer and then assembles a curriculum from scattered YouTube playlists and course catalogues, with no ordering, no prerequisites, and no signal telling them a topic is actually finished. When they fail, nothing adapts: the same material is served again and they stall on the same wall for weeks. On the employer side, recruiters receive hundreds of CVs listing identical keywords with no evidence behind any of them, so junior screening collapses into guesswork and capable graduates are filtered out alongside unprepared ones. The cost is months of misdirected study for the learner and an expensive, low-signal screening funnel for the company.

Your solution

Min 250 characters. Name the mechanism, not the ambition.

Devotopia generates a dependency-ordered curriculum from the learner's target role, current skills and experience, then gates every module behind an adaptive exam. Difficulty rises after two consecutive correct answers and falls after two wrong ones, and the final score is weighted by the difficulty actually faced (easy 1.0, medium 1.5, hard 2.0) rather than by raw question count. Passing at 70% unlocks the

dependent modules and writes a verified skill to the learner's passport; failing injects a shorter remedial module generated from the exact questions that were missed, so a learner is never left stuck. Those verified skills - not self-claimed ones - are what the job-match gap analysis compares against and what recruiters search in the candidate pipeline.

Project overview

Min 200 characters. Reused verbatim in the catalogue.

Devotopia is a career-preparation and recruiting platform. A learner states a target job role, uploads a CV or lists current skills, and receives a visual learning roadmap of modules with hour estimates and prerequisites. They study, sit an adaptive exam per module, and earn verified skills, shield badges and an exportable certificate. Companies use the same data in reverse: they browse pre-vetted candidates ranked by actual exam performance instead of CV keywords, and post jobs against which learners can measure and close their skill gaps. Admins manage users, content and moderation. The entire interface runs in English and Arabic with full RTL layout and light and dark themes.

3. Features and customers

Key features

Min 150 characters. One per line, 6 to 12 features.

AI roadmap generator - turns a target role, current skills and experience into a dependency-ordered module tree with hour estimates and canvas coordinates

Adaptive assessment engine - difficulty rises or falls on two-answer streaks and the score is weighted easy 1.0 / medium 1.5 / hard 2.0

Remedial loop - scoring under 70% injects a shorter module generated from the exact questions the learner missed

CV parsing and AI enhancement - resume PDFs are parsed into a structured profile and experience bullets are rewritten for impact

Skill passport and verifiable shield badges - 10 AWS-style badges with standalone SVG export, backed by exam records

Certificate export - full-page A4 landscape credential with print-specific CSS, issued automatically when every module in a track is passed

Job matching and gap closing - verified skills are compared against job requirements and the missing ones can be injected back into the roadmap in one click

Recruiter pipeline - companies browse and filter candidates scored by real exam performance and completion rate

RAG semantic search - resources, FAQs and job descriptions are embedded and retrieved by cosine similarity from a Qdrant vector store

Multi-provider LLM orchestration - Gemini, Groq, OpenAI and HuggingFace in a fallback chain that degrades to a deterministic mock so the product never hard-fails

Gamification and analytics - daily streaks, 26 achievement definitions, 7/30/90-day activity charts and quiz score trends

Bilingual AR/EN - full Arabic and English across every screen with RTL layout switching, plus light and dark themes

Target customer — main segment

Min 60 characters.

Egyptian computer-science students and fresh graduates preparing for their first developer job. They need an ordered curriculum for a named target role and proof of skill they can put in front of a recruiter, and they use the web app for both.

Target customer — other segments (optional)

One per line.

Career switchers - people moving into tech from another field who need a roadmap that starts from what they already know rather than from zero

Employers and technical recruiters - companies screening juniors who want candidates ranked by verified exam performance instead of CV keywords, using the company dashboard

Bootcamps and training centres - organisations that need measurable per-skill outcomes for a cohort rather than attendance certificates

4. Business model

Revenue model

Min 150 characters.

Two subscriptions, both collected through PayPal with the price resolved server-side from the plan so the client can never send its own amount. Learners pay monthly for Pro Learner - unlimited AI roadmap generation, adaptive exams, CV enhancement and certificate export - while the free tier keeps one roadmap and a limited number of exams. Companies pay monthly for the Company tier, which unlocks the verified-candidate pipeline, candidate search and job postings. Learner revenue is recurring per user; company revenue is recurring per hiring account.

Price point (USD)

The number and the unit.

19.99 USD per learner per month (Pro Learner)

99.99 USD per company per month (Company tier)

Your first 100 customers

Min 150 characters. Real places, real people, real channels.

We start inside our own institution: a single graduating cohort in Alexandria is 100+ students we can demo to in one session, with free Pro accounts for a semester in exchange for usage data and a testimonial. From there we go to the student bodies that already run career-prep tracks - IEEE, GDG on Campus and ACM chapters in Alexandria and Cairo - each reaching several hundred students per event. For the company side we approach the Alexandria and Smart Village software houses that already hire from those cohorts, offering the verified-candidate pipeline free for three months in exchange for a reference.

5. Competition and market

Competitor 1 — name

Coursera

Competitor 1 — website

<https://www.coursera.org>

Competitor 1 — what do they do?

A global course marketplace selling university and industry courses, professional certificates and guided projects, with a separate hiring product for employers.

Competitor 1 — how are you different?

Min 100 characters.

Coursera sells a catalogue: the learner picks the courses and the completion certificate proves attendance, not ability, because the final assessment is identical for everyone and can be retaken until it passes. Devotopia generates the curriculum itself from a target role and its prerequisites, and gates each module behind an exam whose difficulty moves with the learner's answers; failing rewrites the roadmap with a remedial module built from the exact questions missed. A recruiter searching our pipeline filters on skills passed at a measured difficulty weight - a signal a Coursera certificate does not carry. Against roadmap.sh, which publishes static community roadmaps, ours is generated per learner and is verified rather than self-ticked.

Which market are you entering?

MENA region

6. Artificial intelligence

Does your project use AI in any form?

Yes

Which of these describe your project?

Not ticked: Agentic, Computer vision, Classical ML.

AI feature — a model is called for one task and returns a result

AI core — a pipeline (RAG, embeddings, vector search, fine-tuning) the main product depends on

Speech — transcription, synthesis or voice interaction

What information does the AI receive?

Min 80 characters.

Target job role, current skill list, education level and years of experience; the parsed CV (personal summary, experience bullets, education, projects, skill tags); the module title, description and topic list being assessed; the learner's answer history for the live session (question text, chosen answer, correctness, difficulty label, time taken); the topics missed in a failed exam; and, for semantic search, the embedded text of learning resources, FAQs and job descriptions.

What does the AI produce?

Min 80 characters.

A roadmap as structured JSON - an array of modules, each with id, title, description, difficulty, estimated hours, topic list, prerequisite module ids and x/y canvas coordinates; quiz batches of five multiple-choice questions with options, correct-answer index, difficulty label and explanation; a shorter remedial module scoped to the missed topics; rewritten CV experience bullets; chat replies grounded in retrieved context; embedding vectors for cosine-similarity search; and synthesised speech for voice notes.

Which AI models or APIs does the project call?

Name them exactly.

Gemini 2.5 Flash (primary roadmap and quiz generation), Gemini 2.0 Flash, Google text-embedding-004, Groq Llama 3.3 70B Versatile (fallback), OpenAI GPT-4o mini, OpenAI text-embedding-3-small, OpenAI tts-1, meta-llama/Llama-3.1-8B-Instruct via HuggingFace, and AssemblyAI for speech-to-text. Qdrant is the vector store behind the retrieval step.

Estimated AI running cost, USD per month at 1,000 users

200

7. Automated decision loop

Does your system decide its own next steps and call tools without a human choosing which?

No penalty for answering No. There is a penalty for answering Yes and then not being able to describe the loop.

No

What does the system decide on its own?

Min 100 characters.

Note: the previous question is answered No. What follows describes the automated decision loop the server runs with no human in it — the branches are chosen by application code, and the model only fills in content. After an exam is submitted, the system decides by itself what the learner's curriculum looks like next: the difficulty the next question is drawn at, whether the module is passed, which dependent modules unlock, whether a remedial module is generated and which topics it covers, and whether a track certificate is issued. Nobody routes any of it and the learner is never asked.

Which tools or APIs does it call by itself?

Min 40 characters.

`generateQuiz(module, difficulty)` — asks the LLM provider chain for the next batch of five questions at the difficulty the engine selected

`generateRemedialNode(missedTopics)` — generates a shorter remedial module scoped to the questions that were failed

`unlockNextRoadmapModules(userId, moduleId)` — walks the prerequisite graph in MongoDB and flips dependent modules out of locked

`checkAndIssueCertification(userId)` — issues the track credential once every module in the track is completed

`recordTopicResult(userId, topicId, passed, score)` — recomputes `fail_percentage` per topic and enqueues a remedial job on the BullMQ queue past the 30% threshold

`streakService.recordActivity(userId)` — updates the daily streak and emits `exam.completed` / `track.completed` events to the achievement listeners

`ragService.retrieveResources(topic) / retrieveJobs(skills)` — cosine-similarity search in Qdrant to ground resource and job recommendations

Describe one full loop, start to finish

Min 100 characters.

A learner submits the fifth answer of a module exam. The server scores each answer by the difficulty actually faced (easy 1.0, medium 1.5, hard 2.0) and gets 62%. That is below the 70% threshold, so the module is marked failed. It collects the questions answered incorrectly and their topics, recomputes `fail_percentage` for the topic, and past the 30% threshold enqueues a remedial job. The job calls the LLM provider chain to generate a shorter module covering only those topics and inserts it into the roadmap ahead of the failed one, then records the streak and emits `exam.completed`. One LLM call and four service calls, no human involved. On a pass the same loop instead unlocks the dependent modules and checks whether the track certificate is now due.

What happens when it gets it wrong?

Min 60 characters.

Every model call runs through a provider fallback chain — OpenAI, Gemini, Groq, HuggingFace, then a deterministic MockProvider — so a provider that errors or returns nothing falls through to the next one rather than to the user. The reply is shape-checked before it is accepted: a roadmap without a non-empty modules array is discarded and the next provider is tried. Answer grading compares against both the string value and the numeric option index, so a model that returns the wrong shape still grades correctly. If every provider fails, the mock returns a valid hand-written structure and the product keeps working offline. The difficulty and remedial rules are pure functions covered by unit tests, so a wrong decision there is a test failure rather than a runtime surprise.

Why does this need to be agentic?

Min 150 characters.

It does not need to be, and we did not build it that way — the honest answer to the previous question is No. The decisions here are few and enumerable: three difficulty levels, one pass threshold at 70%, one remedial threshold at 30%, and a prerequisite graph that is already explicit in the data. A fixed sequence expresses all of that completely, and it is cheaper, faster and testable — the difficulty ladder and the remedial trigger are covered by unit tests, which would not be possible if a model were choosing the path each run. We spent the model on what genuinely varies between learners, which is the content of the curriculum and the questions, and kept the control flow deterministic. Building more agency than this problem needs would have cost us the determinism the assessment logic depends on.

8. Before you submit

1. First 100 customers: the route described is plausible but no real customer is named. The form states that one sentence about someone you have already spoken to outweighs everything else in that answer. Add it if it exists.
2. The agentic question is answered No, and the five follow-ups describe the real server-side loop while stating plainly that the branches are chosen by application code and the model only fills in content. Keep those two consistent if you edit either.
3. Competitor: Coursera was chosen as the largest, best-known player, with roadmap.sh named inside the differentiation answer. A closer Egyptian or MENA competitor would be a stronger choice if one fits.
4. Every figure quoted is traceable to the source: 70% pass threshold and the 1.0/1.5/2.0 difficulty weights, the 30% remedial threshold, 19.99 and 99.99 USD plan prices, and the OpenAI to Gemini to Groq to HuggingFace to Mock provider order.